

Jenkins + ngrok in WSL

Setup steps completed on 2026-09-13. Jenkins container: jenkins. WSL: Ubuntu-24.04.

1. Folder analysis

Root folder contains a Python Docker demo, Jenkins Docker Hub pipeline, Node multibranch demo, local pipeline scripts, and Jenkins Docker Hub guide.

Jenkins runs as Docker container jenkins in WSL. Jenkins UI uses port 8080. Jenkins agent uses port 50000. Demo app can use port 8081.

2. Environment checks

```
wsl --list --verbose
wsl -d Ubuntu-24.04 -- docker ps
```

Checks confirmed Ubuntu-24.04 running, Jenkins container active, no ngrok installed, and sudo password required.

3. Installed ngrok without sudo

Used official ngrok Linux x64 archive and installed it under WSL user directory. Installed version: ngrok 3.39.11.

```
wsl -d Ubuntu-24.04 -- mkdir -p /home/eslam/.local/bin /tmp/ngrok-install
wsl -d Ubuntu-24.04 -- curl -fL --retry 3
https://bin.equinox.io/c/bNyj1mQVY4c/ngrok-v3-stable-linux-amd64.tgz -o
/tmp/ngrok-v3-stable-linux-amd64.tgz
wsl -d Ubuntu-24.04 -- tar -xzf /tmp/ngrok-v3-stable-linux-amd64.tgz -C
/tmp/ngrok-install
wsl -d Ubuntu-24.04 -- install -m 0755 /tmp/ngrok-install/ngrok
/home/eslam/.local/bin/ngrok
wsl -d Ubuntu-24.04 -- /home/eslam/.local/bin/ngrok version
```

4. Configured ngrok auth token

```
wsl -d Ubuntu-24.04 -- /home/eslam/.local/bin/ngrok config add-authtoken
<NGROK_AUTH_TOKEN>
wsl -d Ubuntu-24.04 -- /home/eslam/.local/bin/ngrok config check
```

Token stored only in /home/eslam/.config/ngrok/ngrok.yml. Token not written to Jenkins folder or repository.

5. Started Jenkins tunnel

```
wsl -d Ubuntu-24.04 -- setsid -f /home/eslam/.local/bin/ngrok http http://localhost:8080
--log=/home/eslam/ngrok-jenkins.log --log-format=json
wsl -d Ubuntu-24.04 -- curl -sS http://127.0.0.1:4040/api/tunnels
```

Active public tunnel: <https://jaybird-rare-mastiff.ngrok-free.dev>. Local target: <http://localhost:8080>.

6. Updated Jenkins public URL

```
wsl -d Ubuntu-24.04 -- docker exec jenkins cp
/var/jenkins_home/jenkins.model.JenkinsLocationConfiguration.xml
/var/jenkins_home/jenkins.model.JenkinsLocationConfiguration.xml.bak-ngrok
wsl -d Ubuntu-24.04 -- docker exec jenkins sh -c "sed -i
's#http://localhost:8080/#https://jaybird-rare-mastiff.ngrok-free.dev/#g'
```

```
/var/jenkins_home/jenkins.model.JenkinsLocationConfiguration.xml"
wsl -d Ubuntu-24.04 -- docker restart jenkins
```

Jenkins URL changed from <http://localhost:8080/> to <https://jaybird-rare-mastiff.ngrok-free.dev/>. Config backup retained inside Jenkins container.

7. Verified tunnel and Jenkins

```
wsl -d Ubuntu-24.04 -- curl -sS -o /dev/null -w "LOCAL %{http_code}"
"http://localhost:8080/login"
wsl -d Ubuntu-24.04 -- curl -sS -o /dev/null -w "NGROK %{http_code}"
"https://jaybird-rare-mastiff.ngrok-free.dev/login"
```

Local Jenkins login returned HTTP 200. Public ngrok Jenkins login returned HTTP 200.

8. Webhook status

GitHub plugin installed. Existing jobs had no GitHub push trigger. Generic webhook endpoint available: <https://jaybird-rare-mastiff.ngrok-free.dev/github-webhook/>.

Jenkins authentication remains enabled. ngrok free URL changes on tunnel restart unless reserved domain configured.